

Outline

- Introduction
 - Background
 - O-RAN architecture
 - A1 interface
 - E2 interface
 - ML Workflow with RIC: Non-RT RIC
 - ML Workflow with RIC: Near-RT RIC
 - AI/ML Models in O-RAN Use Cases
- O-RAN implementation guide on ubuntu 20.04
 - Install k8s, helm, and docker
 - Install Near-RT RIC
 - Install Non-RT RIC
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Introduction (1/14)

A. Background

- Today, radio access network (RAN) components are monolithic units, *all-in-one solutions* that implement each and every layer of the cellular protocol stack.
- They are provided by a limited number of vendors and seen by the operators as black-boxes.
- Reliance on black-box solutions has resulted in:
 - Limited reconfigurability of the RAN, with equipment whose operations cannot be fine-tuned to support diverse deployments and different traffic profiles.
 - Limited coordination among network nodes, preventing joint optimization and control of RAN components.
 - Vendor lock-in, with limited options for operators to deploy and interface RAN equipment from multiple vendors.
- To overcome these limitations, in the last decade several research and standardization efforts have promoted the **Open RAN** (O-RAN) as the new paradigm for the RAN of the future.
- O-RAN deployments are based on **disaggregated**, **virtualized** and **software-based** components, connected through open and well-defined interfaces, and interoperable across different vendors.
 - Disaggregation and virtualization enable flexible deployments.
 - Open and interoperable interfaces also allow operators to onboard different equipment vendors.
 - Open interfaces and software-defined protocol stacks enable the integration of intelligent, data-driven closed-loop control for the RAN.

Introduction (2/14)

A. Background

O-RAN ALLIANCE

– Who we are

- O-RAN ALLIANCE is a world-wide community of mobile operators, vendors, and research & academic institutions with the mission to re-shape Radio Access Networks to be more **intelligent, open, virtualized** and **fully interoperable**.
- Membership: AT&T, NTT DOCOMO, SK telecom, 中華電信,

– What We Do

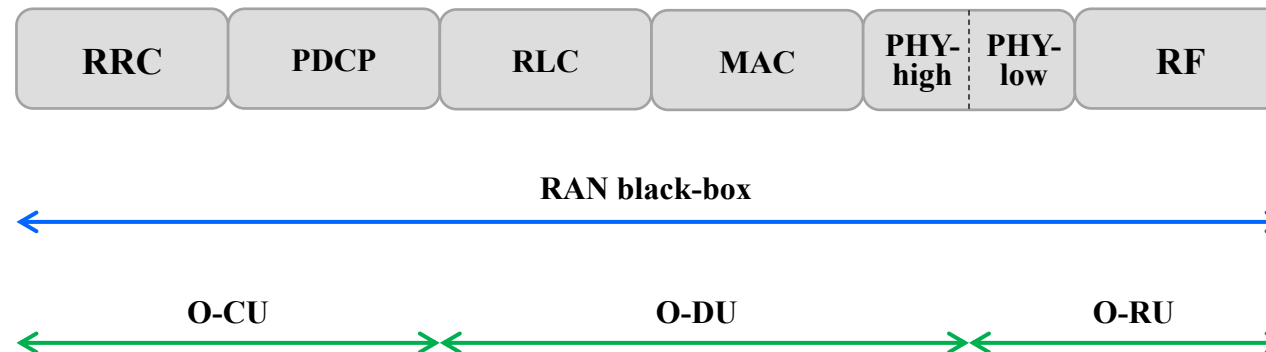
- Specification Effort
 - O-RAN specifications define individual parts of the O-RAN Architecture.
- Open Software for the RAN
 - The O-RAN Software Community aims to create open software for Radio Access Network (RAN), aligning it with O-RAN Architecture and specs.
 - The goal is to unify and speed up RAN evolution and deployment.
- Testing & Integration of O-RAN Implementations
 - O-RAN ALLIANCE aids members in testing and integrating their O-RAN implementations.
 - Open Testing and Integration Centres (OTIC) provide a collaborative, open, and impartial testing environment.
- O-RAN Demonstrations
 - O-RAN Virtual Exhibition allows companies to demonstrate their O-RAN based implementations.



Introduction (3/14)

B. O-RAN architecture

- The gNB is split into a Central Unit (CU), a Distributed Unit (DU), and a Radio Unit (RU) (called O-CU, O-DU, and O-RU in O-RAN specifications).
 - The **O-RU** performs time-domain functionalities, with precoding, Fast Fourier Transform (FFT), cyclic prefix addition/removal, and Radio Frequency (RF) operations.
 - The **O-DU** then takes care of the remaining functionalities of the physical (PHY) layer, and of the Medium Access Control (MAC) and Radio Link Control (RLC) layers.
 - scrambling, modulation, layer mapping, part of precoding, and mapping into physical resource blocks
 - The **O-CU** units (CP and UP) implement the higher layers of the 3GPP stack.
 - Radio Resource Control (RRC) layer: manages the life cycle of the connection
 - Service Data Adaptation Protocol (SDAP) layer: manages the Quality of Service (QoS) of the traffic flows
 - Packet Data Convergence Protocol (PDCP) layer: takes care of reordering, packet duplication, and encryption for the air interface



Introduction (4/14)

B. O-RAN architecture

- The second innovation is represented by the RAN intelligent controllers (RICs), which introduce programmable components that can run closed-loop control and orchestrate the RAN.
 - **Non-real-time RIC:**
 - Provides guidance, enrichment information, and management of machine learning (ML) models for the Near-RT RIC.
 - The AI algorithms run as rApps on a time scale larger than 1 second.
 - It has been designed to derive control policies that operate at a **higher level** and affect a larger number of users and network nodes.
 - **Near-real-time RIC:**
 - Deliver specific services such as radio resource management (RRM) and handover controlling for network.
 - AI algorithms are implemented as external applications (i.e., xApps) and operate control loops with a periodicity between 10 ms and 1 s.
- The O-RAN Alliance has also introduced technical specifications that describe open interfaces connecting a number of different components of the O-RAN architecture.
 - **O1:** SMO to each O-RAN component
 - **A1:** Non-RT RIC to Near-RT RIC
 - **E2:** Near-RT RIC to each E2 nodes (O-CU, O-DU)

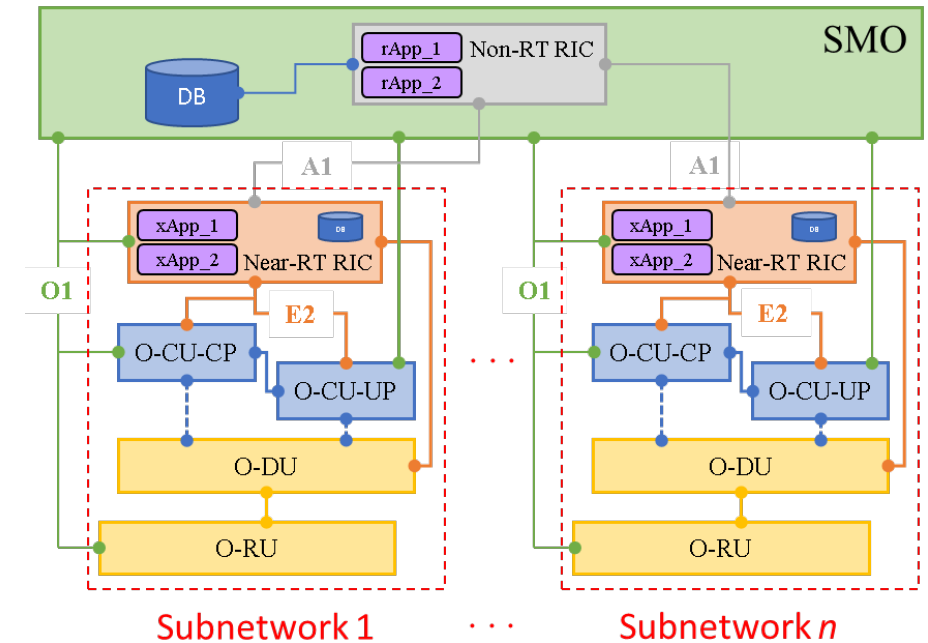


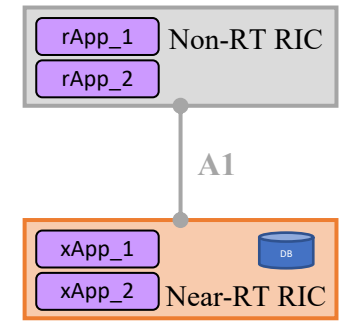
Fig. O-RAN architecture

SMO: Service Management and Orchestration

Introduction (5/14)

C. A1 interface

- The A1AP (A1 application protocol) has three services for near-RT RIC including *A1 Policy management*, *A1 ML model management* and *A1 Enrichment Information*.
 - **A1 Policy management:**
 - To drive the functionalities of the near-RT RIC to achieve high level intent for the RAN.
 - This intent is generally defined through **QoS or KPI goals** for all users or subsets of users and **monitored** using the O1 interface and feedback over A1 to near-RT RIC.
 - Policies, formulated by the non-RT RIC, are deployed via A1, which also enables policy lifecycle management, including **deletion**, **updates**, and **queries**, by the non-RT RIC.
 - **A1 ML model management:** Be considered as further study.
 - **A1 Enrichment Information (EI):**
 - Aims at improving RAN performance by providing information that is generally not available to the RAN itself (e.g., capacity forecasts, information elements from sources outside the RAN, aggregate analytics).
 - The non-RT RIC and SMO possess a network-wide view and external data access, and can share the data with near-RT RIC xApps via A1 EI.
 - The flow of information can also bypass the non-RT RIC, which can instruct the near-RT RIC (over A1) to connect directly to sources of EI.



QoS: quality of service

Introduction (6/14)

C. A1 interface

- An example scenario used to describe the use of A1 for traffic management.
- In this scenario, an UE with **UE_ID = 1**, belonging to a subnet slice identified by **S-NSSAI = 1**, having a **Voice (5QI = 1)** and an **MBB (5QI = 9)** connection established, enters an area covered by four frequency bands.

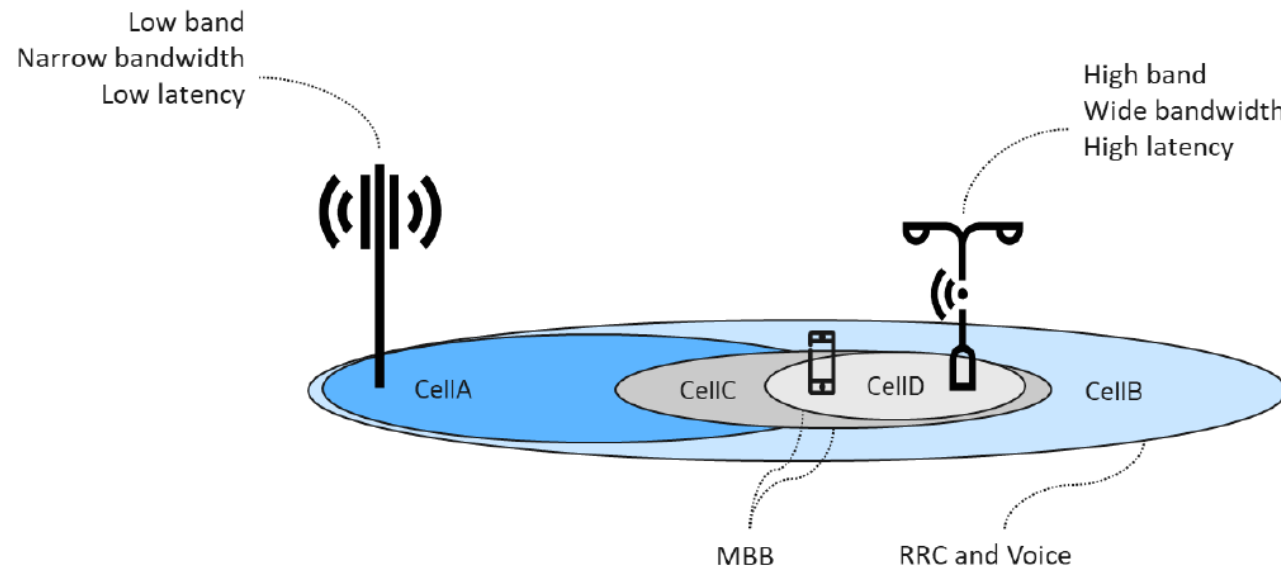


Figure 3.1.5-1: Desired use of the cells

MBB: Mobile broadband

S-NSSAI: Single Network Slice Selection Assistance Information

Introduction (7/14)

C. A1 interface

- The Non-RT RIC understands the requirements and characteristics of the services and decides to let the **Voice** and **RRC connection** reside on the **low band** (CellB: narrow bandwidth).

```
{
  "policy_id": "1",
  "scope": {
    "ue_id": "1",
    "slice_id": "1",
    "qos_id": "1",
    "cell_id": "X" // Policy for Cell X, where X is one of A, B, C or D
  },
  "statement": {
    "cell_id_list": "B",
    "preference": "Shall",
    "primary": true // Control plane on Cell B (becoming PCell)
  },
  "statement": {
    "cell_id_list": "B",
    "preference": "Shall",
    "primary": false // Voice on Cell B
  }
}
```

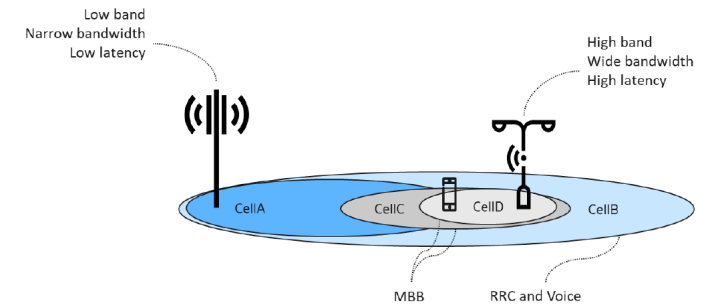


Figure 3.1.5-1: Desired use of the cells

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C. A1 interface

- While the **MBB connection** should preferably use the **higher band** (CellC, CellD: wide bandwidth) and avoid the low band (CellA, CellB) if possible.
 - CellA is used for MBB if required for coverage reasons.

```
}  
  
{  
  "policy_id": "2",  
  "scope": {  
    "ue_id": "1",  
    "slice_id": "1",  
    "qos_id": "9",  
    "cell_id": "X" // Policy for Cell X, where X is one of A, B, C or D  
  },  
  "statement": {  
    "cell_id_list": {"B", "A"},  
    "preference": "Avoid",  
    "primary": false // Avoid MBB on Cell A and Cell B  
  },  
  "statement": {  
    "cell_id_list": {"C", "D"},  
    "preference": "Prefer",  
    "primary": false // Prefer MBB on Cell C and Cell D  
  }  
}
```

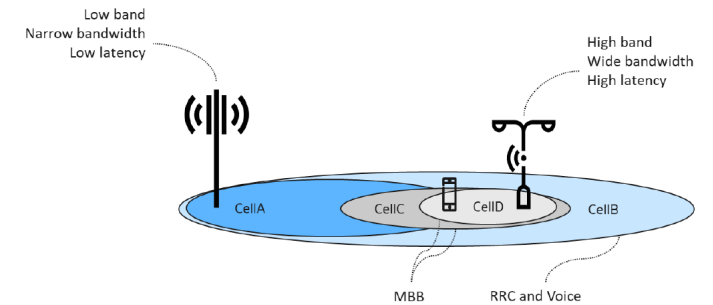
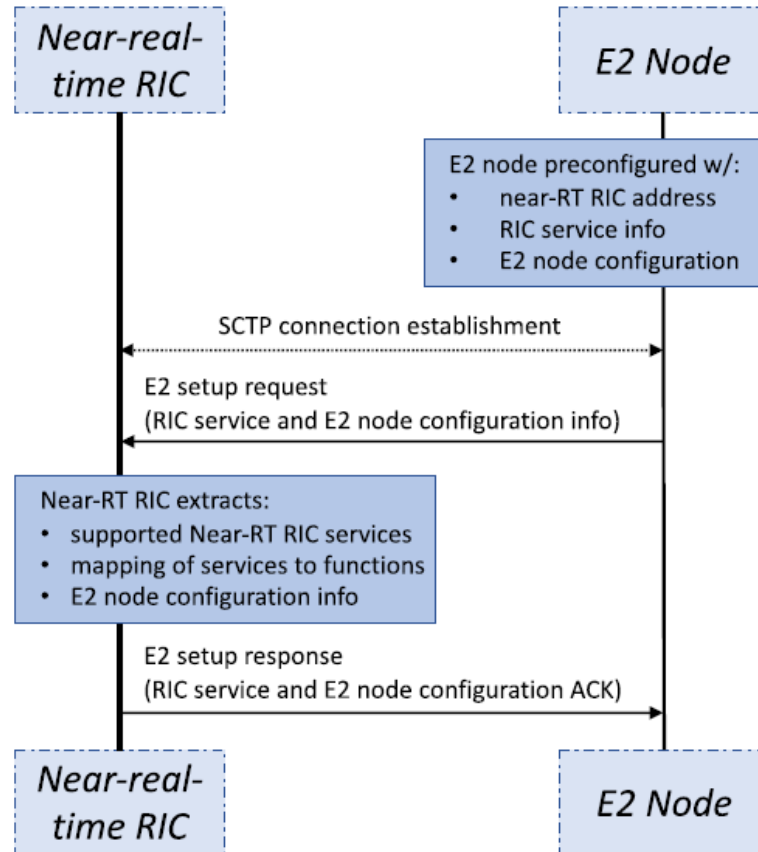


Figure 3.1.5-1: Desired use of the cells

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D. E2 interface



1. At first, the **SCTP connection** is established.
2. Then, the E2 node transmits an **E2 setup request**, in which it lists the RAN functions and configuration it supports, together with the identifiers for the node.
3. Finally, the near-RT RIC processes this information and replies with an **E2 setup response**.

Fig. The procedure of setting up an E2 session between the near-RT RIC and an E2 node.

SCTP: Stream Control Transmission Protocol

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D. E2 interface

- After the connection is established, the E2AP (E2 Application Protocol) provides four services,
 - **E2 Report:**
 - The E2 report service sends data from an E2 node to an xApp in the near-RT RIC.
 - Subscription triggers or periodicity determine when the E2 node sends report messages containing telemetry data.
 - **E2 Insert:**
 - The insert service notifies the near-RT RIC xApp about **specific E2 node events**, such as a potential handover signaled by a UE (user equipment).
 - Upon the E2 node sending the insert message, a wait timer also begins, and if the RIC doesn't respond before it ends, the E2 node procedure can resume or be definitively halted.
 - **E2 Control:**
 - The control service can be initiated by the RIC or triggered by receiving an insert message at the near-RT RIC.
 - It involves two messages: a RIC Control Request from RIC to E2 nodes, and a reverse RIC Control Acknowledge.
 - **E2 Policy:**
 - This service involves a subscription procedure that specifies: (i) an event trigger; and (ii) a policy that the E2 node should autonomously follow to perform radio resource management.
- These services are then combined to create a service model.

Introduction (11/14)

D. E2 interface

- An example of E2 message exchange between an E2 node and a near-RT RIC xApp in a standard control procedure.

E2 report

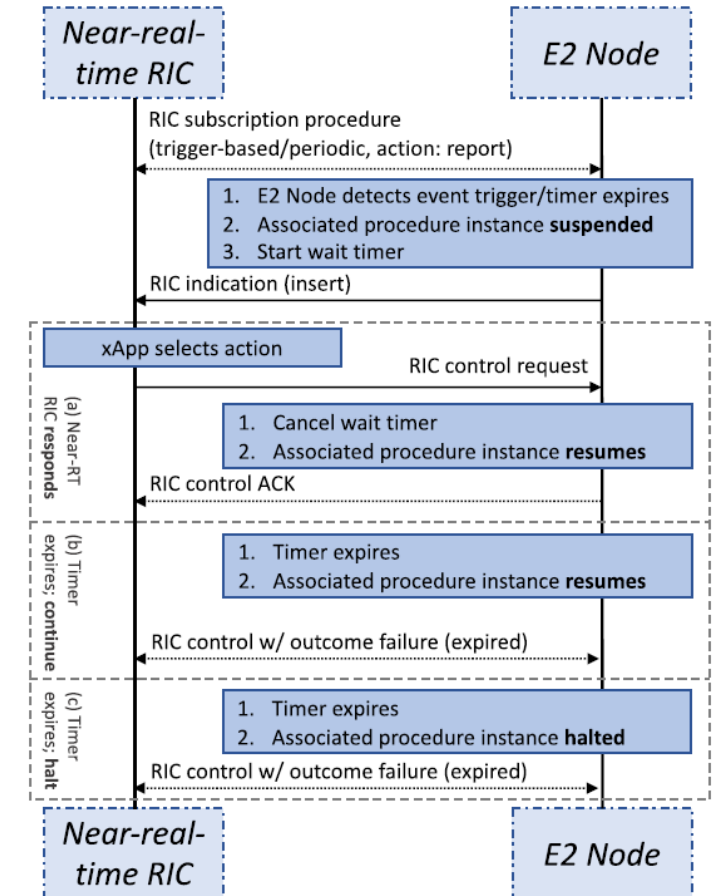
- First, the xApp **subscribes** to the E2 node, which needs to expose a specific RAN function called *RAN control*.
 - During the subscription phase, the xApp can specify a triggering event or a timer.

E2 insert

- Then, if the timer expires or the condition defined in the trigger is verified, the E2 node sends an E2 control request insert message to the RIC.
 - At the same time, the E2 node suspends the radio resource management procedure, and starts a timer.

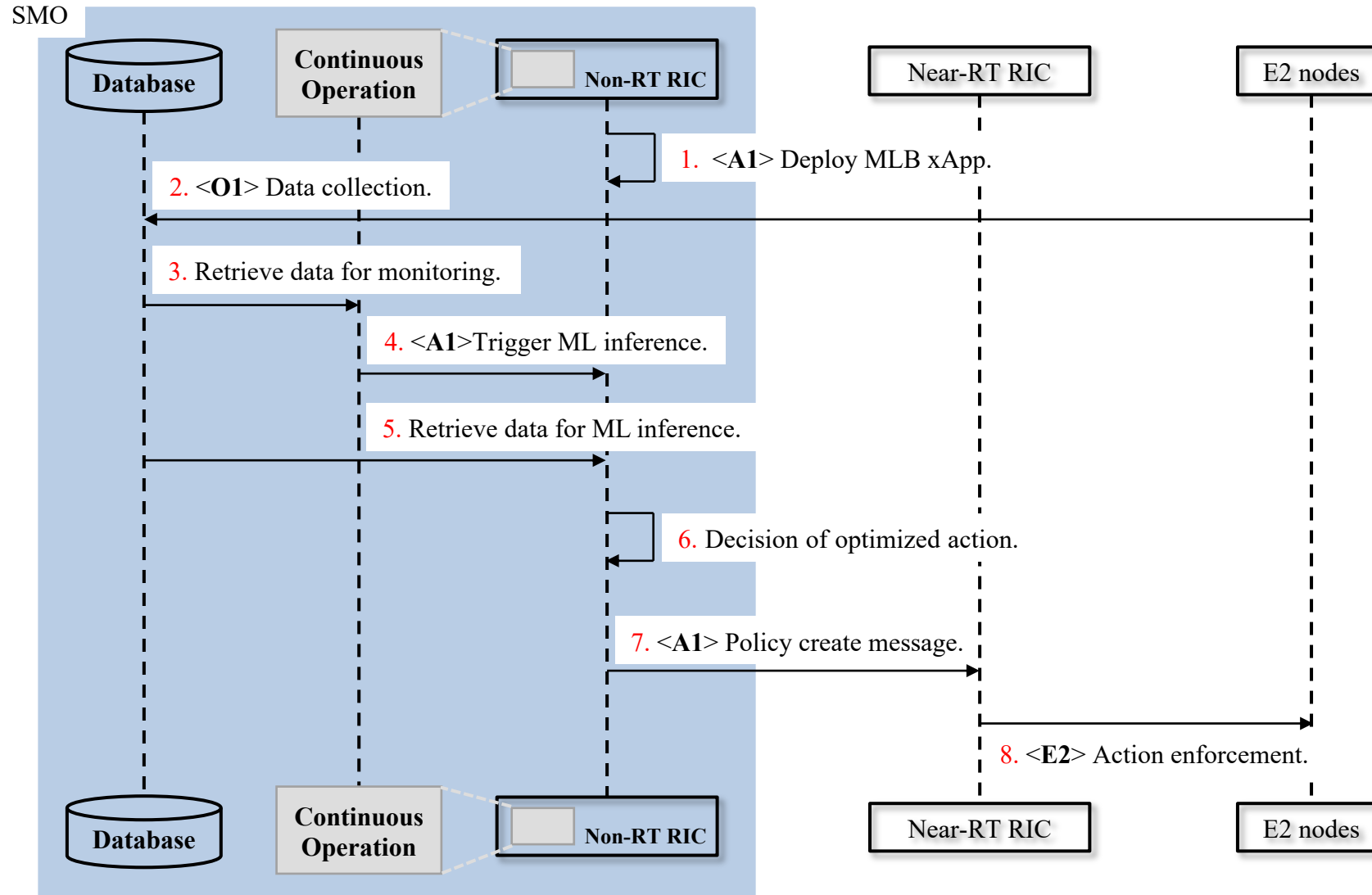
E2 control

- Three different outcomes will follow:
 - The RIC can reply with an E2 control message that **denies** the control request or **accepts** it and replies with a control action.
 - If the E2 node timer expires before the reception of a message from the RIC, the procedure may **continue** autonomously, or it may be **halted**.



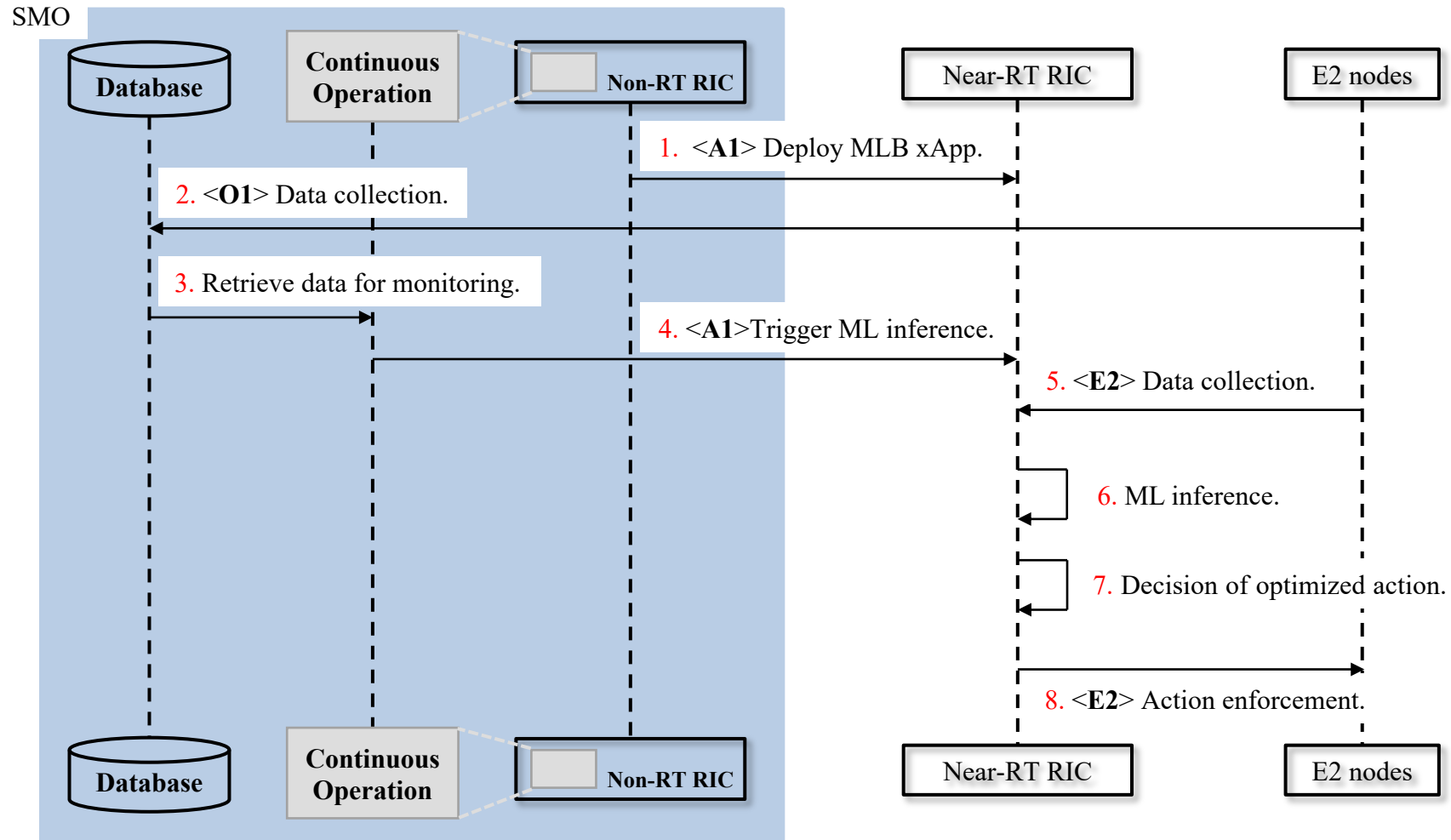
Introduction (12/14)

E. ML Workflow with RIC: Non-RT RIC



Introduction (13/14)

F. ML Workflow with RIC: Near-RT RIC



Introduction (14/14)

G. AI/ML Models in O-RAN Use Cases

Use Case	AI/ML models functionality description	AI/ML algorithms types (example)	Data Input	Data Output
Traffic steering	Cell load prediction/user traffic volume prediction	Supervised learning (time series prediction, e.g., SVR, DNN)	UL/DL PRB occupation	Same to input
	Radio finger print prediction	Supervised learning (e.g., SVR, GBDT)	Intra-frequency MR data and PM counters, e.g., RSRP, RSRQ, MCS, CQI, etc.	Inter-frequency MR data, e.g., RSRP, RSRQ, MCS, CQI, etc.
V2X handover management	- Prediction / detection HO anomalies - Discovery of preferred HO sequences	Supervised learning	- Cam - Radio cell IDs - Connection IDs - Basic radio measurements (RSRP, RSRQ etc.) - GPS - Direction - Velocity	- HO anomalies probability - Preferred HO sequences

MR: measurement report
 RSRP: reference Symbol Received Power
 RSRQ: reference Signal Received Quality
 CQI: channel Quality Indicator
 MCS: modulation and Coding Scheme

DNN: deep neural network
 UL/DL: uplink/downlink
 HO: handover
 PRB: physical Resource Block

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Install k8s, helm, and docker (1/2)

- 首先 clone 兩個不同版本的 ric-dep 資料夾至不同路徑
 - `git clone --branch f-release https://gerrit.o-ran-sc.org/r/ric-plt/ric-dep` 至路徑 1: `/Home/ric-dep`
 - `git clone https://gerrit.o-ran-sc.org/r/ric-plt/ric-dep` 至路徑 2: `/root/ric-dep`
 - 再將路徑 2 的 `install_k8s_and_helm.sh` 檔案取代路徑 1 內同名的檔案
 - 之後我們利用路徑 1 的資料夾安裝 k8s (1.19.11), helm (3.5.4), docker and Near-RT RIC (f-release)
- 進入 `/root/ric-dep/bin/install_k8s_and_helm.sh` 修改 .sh 檔
 - 修改 k8s 的版本至 1.19.11

```
57     echo "" >> /etc/network/interf
58     echo "allow-hotplug ${IPv6IF}"
59     echo "iface ${IPv6IF} inet6 :::
60     ifconfig ${IPv6IF} up
61 fi
62 }
63
64 KUBEV="1.19.11"
65 KUBECNIV="0.7.5"
66 HELMV="3.5.4"
67 DOCKERV="20.10.21"
68
69 echo running ${0}
70 while getopts ":k:d:e:n:c" o; do
```

```
102 rm -rf /opt/config
103 mkdir -p /opt/config
104 echo "" > /opt/config/docker_version.txt
105 echo "1.19.11" > /opt/config/k8s_version.txt
106 echo "0.7.5" > /opt/config/k8s_cni_version.txt
107 echo "3.5.4" > /opt/config/helm_version.txt
108 echo "${hostname -I}" > /opt/config/host_private_
109 echo "$(curl ifconfig.co)" > /opt/config/k8s_mst_
110 echo "${hostname -I}" > /opt/config/k8s_mst_privat
111 echo "mtu" > /opt/config/mtu.txt
112 echo "cinder_volume_id" > /opt/config/cinder_
113 echo "${hostname}" > /opt/config/stack_name.txt
```

```
282 kind: KubeProxyConfiguration
283 mode: ipvs
284 EOF
285 | elif [[ ${KUBEV} == 1.15.* ]] || [[ ${KUBEV} == 1.19.* ]] || [[ ${KUBEV} == 1.18.* ]]; then
286     cat <<EOF >/root/config.yaml
287 apiVersion: kubeadm.k8s.io/v1beta2
288 kubernetesVersion: v${KUBEV}
289 kind: ClusterConfiguration
290 apiServer:
291   extraArgs:
```

Install k8s, helm, and docker (2/2)

- 將網址改成如圖

```
339 kubectl get pods --all-namespaces
340
341 # we refer to version 0.18.1 because later versions use namespace kube-flannel instead of kube-system TODO
342 kubectl apply -f "https://raw.githubusercontent.com/coreos/flannel/master/Documentation/kube-flannel.yml"
343
```

- 將 pods 的數量改為 7 個

```
341 # we refer to version 0.18.1 because later versions use namespace k
342 kubectl apply -f "https://raw.githubusercontent.com/coreos/flannel/i
343
344 wait_for_pods_running 7 kube-system|
345
346 kubectl taint nodes --all node-role.kubernetes.io/master-
347
348 HELM=$(cat /etc/os-release | grep ^ID= | cut -d= -f2 | tr -d ' ' |
```

- cd /root/ric-dep/bin
- ./install_k8s_and_helm.sh
- ./install_common_templates_to_helm.sh
- ./setup-ric-common-template
- Check version:

```
root@TF-BC0121P:~# kubectl version
Client Version: version.Info{Major:"1", Minor:"19", GitVersion:"v1.19.11", GitCommit:"c6a2f8fc437
8c5381dd948d9ad9d1880e3e0b33", GitTreeState:"clean", BuildDate:"2021-05-12T12:27:07Z", GoVersion:"
go1.15.12", Compiler:"gc", Platform:"linux/amd64"}
Server Version: version.Info{Major:"1", Minor:"19", GitVersion:"v1.19.11", GitCommit:"c6a2f8fc437
8c5381dd948d9ad9d1880e3e0b33", GitTreeState:"clean", BuildDate:"2021-05-12T12:19:22Z", GoVersion:"
go1.15.12", Compiler:"gc", Platform:"linux/amd64"}
root@TF-BC0121P:~#
```

```
root@TF-BC0121P:~# helm version
version.BuildInfo{Version:"v3.5.4", GitCommit:"1b5edb69df3d3a88df77c9902dc17af
864ff05d1", GitTreeState:"clean", GoVersion:"go1.15.11"}
root@TF-BC0121P:~#
```

```
root@TF-BC0121P:~# docker version
Client:
Version: 20.10.21
API version: 1.41
Go version: go1.18.1
Git commit: 20.10.21-0ubuntu1-20.04.2
Built: Thu Apr 27 05:56:19 2023
OS/Arch: linux/amd64
Context: default
Experimental: true

Server:
Engine:
Version: 20.10.21
API version: 1.41 (minimum version 1.12)
Go version: go1.18.1
Git commit: 20.10.21-0ubuntu1-20.04.2
Built: Thu Apr 27 05:37:01 2023
OS/Arch: linux/amd64
Experimental: false
containerd:
Version: 1.7.2
GitCommit:
runc:
Version: 1.1.7-0ubuntu1-20.04.1
GitCommit:
docker-init:
Version: 0.19.0
GitCommit:
root@TF-BC0121P:~#
```

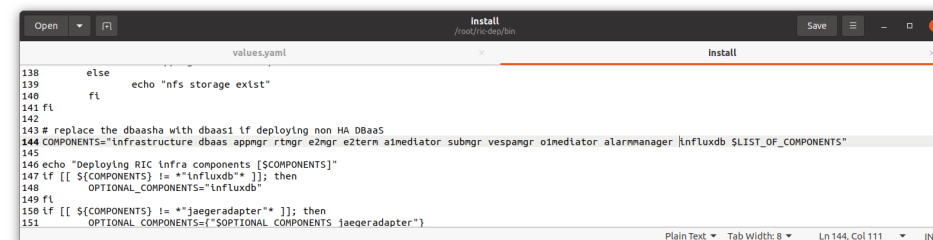
Install Near-RT RIC (1/2)

- 修改兩個檔案內 tiller 的 tag version (如右圖) [2]:

- 檔案1路徑: *ric-dep/helm/appmgr/values.yaml*
- 檔案2路徑: *ric-dep/helm/infrastructure/values.yaml*

```
27 tillerTLSecrets:
28   registry: nexus3.o-ran-sc.org:10002/o-r
29   name: it-dep-secret
30   tag: 0.0.2
31   tiller:
32     registry: ghcr.io
33     name: helm/tiller
34     tag: v2.12.3
35   secret:
36     create: true
37     tillerSecretName: ricxapp-tiller-secret
38     helmSecretName: ricxapp-helm-secret
39   tls:
```

- **kubectl create ns ricinfra**
- **helm repo add stable https://charts.helm.sh/stable**
- **helm install nfs-release-1 stable/nfs-server-provisioner --namespace ricinfra**
- **kubectl patch storageclass nfs -p '{"metadata": {"annotations":{"storageclass.kubernetes.io/is-default-class":"true"}}}'**
- **apt install nfs-common**
- add *influxdb* component into the line "COMPONENTS" in the file *dep/ric-dep/bin/install*



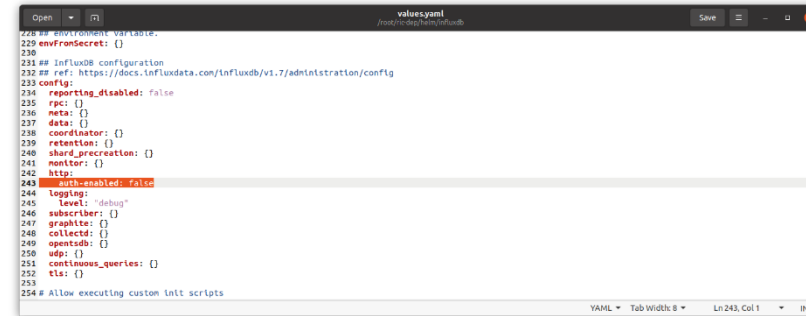
```
138 else
139   fi echo "nfs storage exist"
140   fi
141 fi
142
143 # replace the dbaas with dbaas1 (if deploying non HA DBaaS)
144 COMPONENTS="infrastructure dbaas appmgr rtmgr e2mgr e2term ainediator submgr vespangr oinediator alarmmanager influxdb $LIST_OF_COMPONENTS"
145
146 echo "Deploying RIC infra components [$COMPONENTS]"
147 if [[ $(COMPONENTS) != ""influxdb"" ]]; then
148   OPTIONAL_COMPONENTS="influxdb"
149 fi
150 if [[ $(COMPONENTS) != ""jaegeradapter"" ]]; then
151   OPTIONAL_COMPONENTS="$OPTIONAL_COMPONENTS jaegeradapter"
```

Install Near-RT RIC (2/2)

- 當 xApp 欲利用 influxDB 的資源時，若未設定授權選項，會出現以下字樣，

```
"Failed to connect to influxdb: 401: {\\"error\\":\\"authorization failed\\"}"}
```

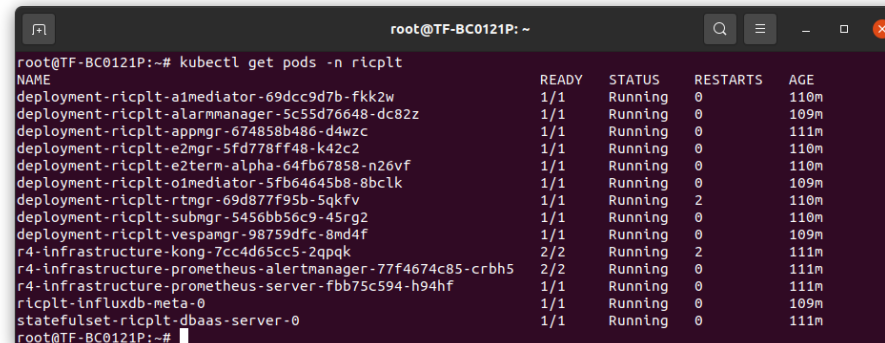
- 因此需進入 `ric-dep/helm/influxdb/values.yaml`，修改 `auth-enabled: true` 至 `false` 以排除問題



```
228 # environment variable.
229 envFromSecret: {}
230
231 ## InfluxDB configuration
232 ## ref: https://docs.influxdata.com/influxdb/v1.7/administration/config
233 config:
234   reporting_disabled: false
235   rpc: {}
236   meta: {}
237   data: {}
238   coordinator: {}
239   retention: {}
240   shard_precreation: {}
241   monitor: {}
242   http:
243     auth-enabled: false
244   logging:
245     level: "debug"
246   subscriber: {}
247   graphite: {}
248   collectd: {}
249   opentsdb: {}
250   udp: {}
251   continuous_queries: {}
252   tls: {}
253
254 # Allow executing custom init scripts
```

- 安裝 RIC platform (Near-RT RIC)

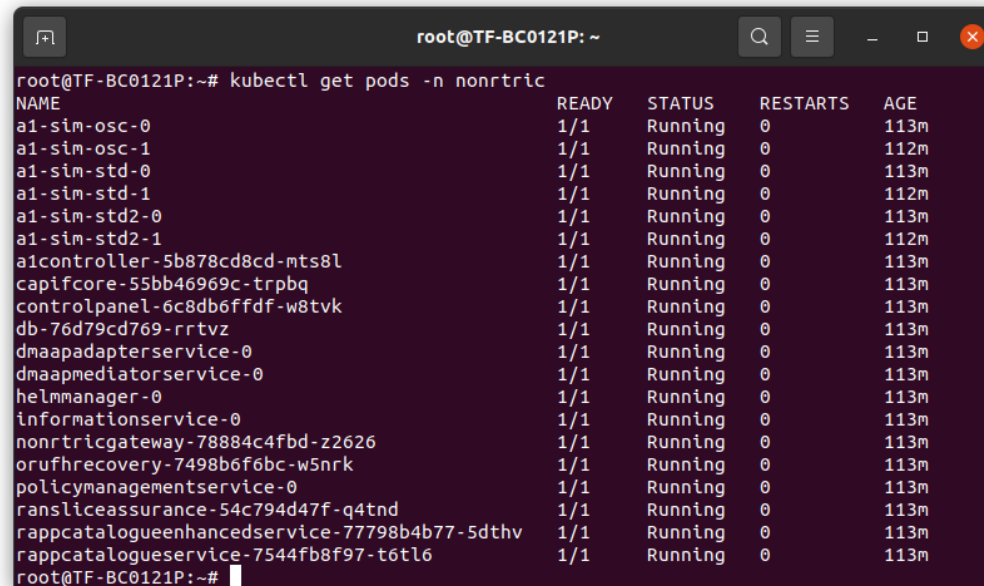
- `./install -f ../RECIPE_EXAMPLE/example_recipe_latest_stable.yaml`
- `kubectl get pods -n ricplt %` 查看 pods 狀態



```
root@TF-BC0121P:~# kubectl get pods -n ricplt
NAME                                READY   STATUS    RESTARTS   AGE
deployment-ricplt-a1mediator-69dcc9d7b-fkk2w   1/1     Running   0           110m
deployment-ricplt-alarmmanager-5c55d76648-dc82z 1/1     Running   0           109m
deployment-ricplt-appmgr-074858b486-d4wzc      1/1     Running   0           111m
deployment-ricplt-e2mgr-5fd778ff48-k42c2       1/1     Running   0           110m
deployment-ricplt-e2term-alpha-64fb67858-n26vf 1/1     Running   0           110m
deployment-ricplt-o1mediator-5fb64645b8-8bclk   1/1     Running   0           109m
deployment-ricplt-rtmgr-69d877f95b-5qkfz       1/1     Running   2           110m
deployment-ricplt-submgr-5456bb56c9-45rg2      1/1     Running   0           110m
deployment-ricplt-vespamgr-98759dfc-8md4f      1/1     Running   0           109m
r4-infrastructure-kong-7cc4d65cc5-2qpqk        2/2     Running   2           111m
r4-infrastructure-prometheus-alertmanager-77f4674c85-crbh5 2/2     Running   0           111m
r4-infrastructure-prometheus-server-fbb75c594-h94hf 1/1     Running   0           111m
ricplt-influxdb-meta-0                       1/1     Running   0           109m
statefulset-ricplt-dbaas-server-0             1/1     Running   0           111m
root@TF-BC0121P:~#
```

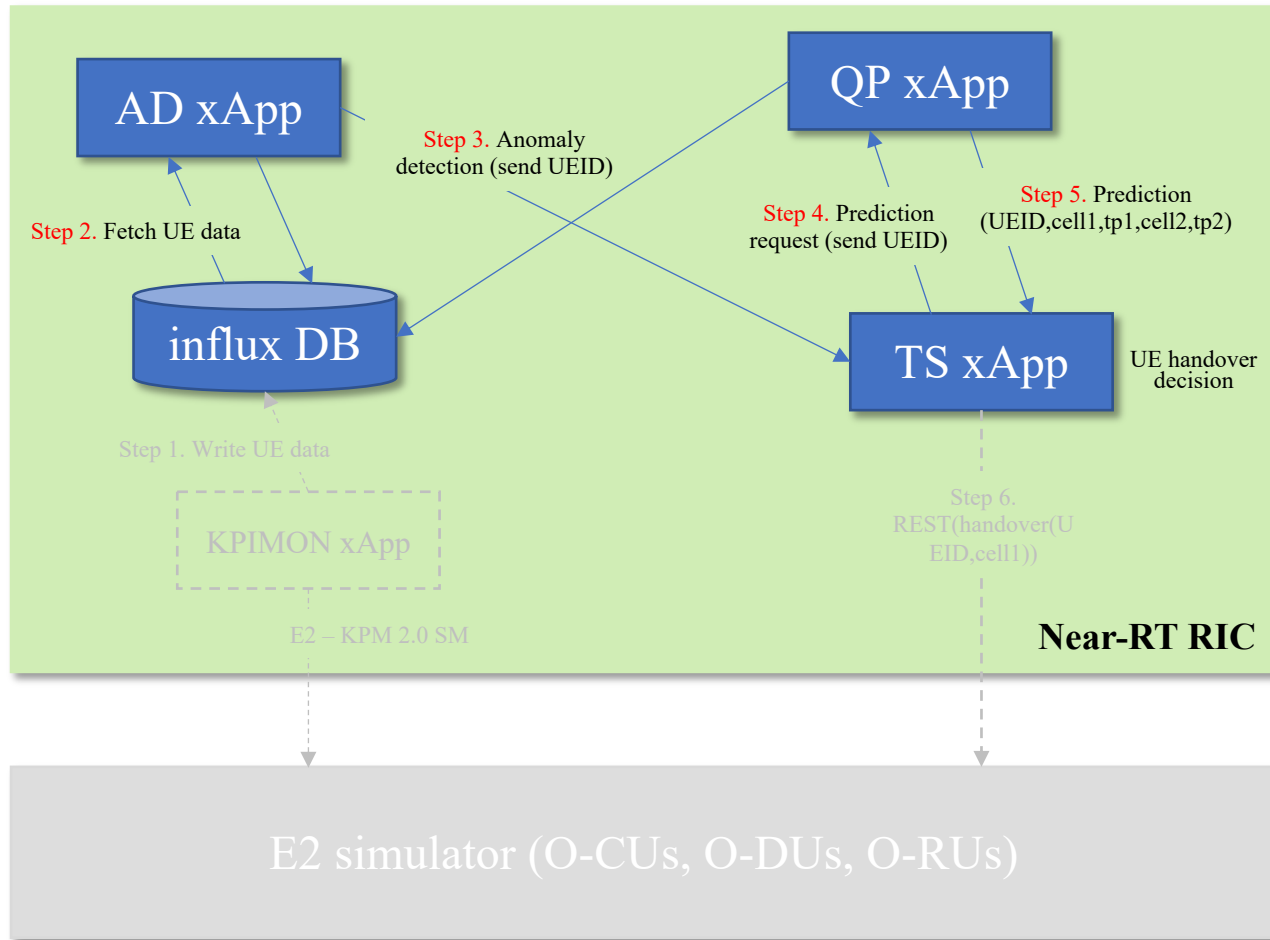
Install Non-RT RIC

- `git clone https://gerrit.o-ran-sc.org/r/it/dep`
- `dep/bin/deploy-nonrttric -f dep/nonrttric/RECIPE_EXAMPLE/example_recipe.yaml` % 安裝 Non-RT RIC
- `kubectl get pods -n nonrttric` % 查看 pods 狀態



```
root@TF-BC0121P: ~  
root@TF-BC0121P:~# kubectl get pods -n nonrttric  
NAME                                READY   STATUS    RESTARTS   AGE  
a1-sim-osc-0                        1/1     Running   0           113m  
a1-sim-osc-1                        1/1     Running   0           112m  
a1-sim-std-0                        1/1     Running   0           113m  
a1-sim-std-1                        1/1     Running   0           112m  
a1-sim-std2-0                      1/1     Running   0           113m  
a1-sim-std2-1                      1/1     Running   0           112m  
a1controller-5b878cd8cd-mts8l      1/1     Running   0           113m  
capifcore-55bb46969c-trpbq         1/1     Running   0           113m  
controlpanel-6c8db6ffdf-w8tvk      1/1     Running   0           113m  
db-76d79cd769-rtrtvz              1/1     Running   0           113m  
dmaapadapterservice-0              1/1     Running   0           113m  
dmaapmediatorservice-0             1/1     Running   0           113m  
helmmanager-0                     1/1     Running   0           113m  
informationservice-0               1/1     Running   0           113m  
nonrttricgateway-78884c4fbd-z2626  1/1     Running   0           113m  
orufhrecovery-7498b6f6bc-w5nrk     1/1     Running   0           113m  
policymanagementservice-0          1/1     Running   0           113m  
ransliceassurance-54c794d47f-q4tnd 1/1     Running   0           113m  
rappcatalogueenhancedservice-77798b4b77-5dthv 1/1     Running   0           113m  
rappcatalogueservice-7544fb8f97-t6tl6 1/1     Running   0           113m  
root@TF-BC0121P:~#
```

Anomaly Detection Use Case (1/9)



- **AD: Anomaly Detection**
- **QP: QoE Predictor**
- **TS: Traffic Steering**
- **DB: Database**
- **xApp deployment sequence: AD→TS→QP**
 - **xApp 佈署 3 步驟:**
 1. 取得 xApp
 2. 上架 xApp
 3. 安裝 xApp

QoE: quality of experience

Anomaly Detection Use Case (2/9)

- Install dms_cli tool:

- `git clone "https://gerrit.o-ran-sc.org/r/ric-plt/appmgr"`
- `cd appmgr/xapp_orchestrator/dev/xapp_onboarder`
- `apt install -y python3-pip`
- `pip3 uninstall xapp_onboarder`
- `pip3 install ./`
- `export CHART_REPO_URL=http://0.0.0.0:8090`
- `dms_cli health % check dms health`

- `docker run --rm -u 0 -it -d -p 8090:8080 -e DEBUG=1 -e STORAGE=local -e STORAGE_LOCAL_ROOTDIR=/charts -v $(pwd)/charts:/charts chartmuseum/chartmuseum:latest`

- Install ad xApp:

- `git clone --branch f-release https://gerrit.o-ran-sc.org/r/ric-app/ad`
- modify the host IP (influxdb service IP) in *database.py* and *insert.py* file

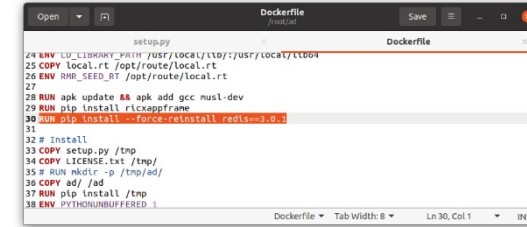
```
root@TF-BC0121P:~# kubectl get svc -n ricplt
NAME                                TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
aux-entry                           ClusterIP    10.109.122.35   <none>            80/TCP,443/TCP   3h49m
r4-infrastructure-kong-proxy         ClusterIP    10.111.117.83   <none>            32088:32088/TCP,32443:32443/TCP   3h49m
r4-infrastructure-prometheus-alertmanager ClusterIP    10.97.169.135   <none>            80/TCP            3h49m
r4-infrastructure-prometheus-server ClusterIP    10.107.90.37    <none>            80/TCP            3h49m
ricplt-influxdb                     ClusterIP    10.97.108.00    <none>            8086/TCP,8088/TCP 3h46m
service-ricplt-aimedator-http        ClusterIP    10.100.170.10   <none>            10080/TCP          3h47m
service-ricplt-aimedator-rnr         ClusterIP    10.96.144.184   <none>            4561/TCP,4562/TCP 3h47m
service-ricplt-alarmmanager-http     ClusterIP    10.110.158.82   <none>            8080/TCP            3h46m
service-ricplt-alarmmanager-rnr       ClusterIP    10.100.149.115   <none>            4560/TCP,4561/TCP 3h46m
service-ricplt-appmgr-http           ClusterIP    10.110.46.210   <none>            8080/TCP            3h48m
service-ricplt-appmgr-rnr             ClusterIP    10.104.107.170   <none>            4561/TCP,4560/TCP 3h48m
service-ricplt-bbaas-tcp              ClusterIP    10.100.149.115   <none>            6379/TCP            3h46m
service-ricplt-e2mgr-http            ClusterIP    10.102.38.237    <none>            3800/TCP            3h48m
service-ricplt-e2mgr-rnr              ClusterIP    10.96.65.642    <none>            4561/TCP,3801/TCP 3h48m
service-ricplt-e2term-prometheus-alpha ClusterIP    10.97.166.129    <none>            8080/TCP            3h47m
service-ricplt-e2term-rnr-alpha       ClusterIP    10.104.138.181   <none>            4561/TCP,3800/TCP 3h47m
service-ricplt-e2term-sctp-alpha      NodePort     10.96.190.33     <none>            34422:32222/SCTP   3h47m
service-ricplt-oiemedator-http        ClusterIP    10.96.40.113     <none>            9001/TCP,8080/TCP,3000/TCP 3h46m
service-ricplt-oiemedator-tcp-netconf NodePort     10.103.95.133     <none>            8301:30830/TCP     3h46m
service-ricplt-rtmgr-http             ClusterIP    10.96.244.201    <none>            3800/TCP            3h48m
service-ricplt-rtmgr-rnr              ClusterIP    10.99.80.157     <none>            4561/TCP,4560/TCP 3h48m
service-ricplt-submgr-http            ClusterIP    None              <none>            3800/TCP            3h47m
service-ricplt-submgr-rnr             ClusterIP    None              <none>            4560/TCP,4561/TCP 3h47m
service-ricplt-vespmgr-http           ClusterIP    10.105.234.226   <none>            8080/TCP,9095/TCP 3h47m
```

```
46 Attributes
47 -----
48 client: InfluxDB client
49 data: DataFrame
50 data: DataFrame
51 ---
52 fetched data from database
53 ---
54
55 def __init__(self, dbname, user='root', password='root', host='10.97.108.00', port='8086'):
56     self.data = None
57     self.client = DataFrameClient(host, port, user, password, dbname)
58
59 def read_data(self, meas, limit=100000):
60     """Read data Method for a given measurement and limit
61     Parameters
62     meas: str (default='ueMeasureReport')
63     limit: int (default=100000)
64
65     result = self.client.query('select * from ' + meas + ' limit ' + str(limit))
66     print('Querying data: ' + meas + ' : size: ' + str(len(result[meas])))
67     try:
68         if len(result[meas]) != 0:
69             self.data = result[meas]
70     except:
```

```
24 import datetime
25
26 class INSERTDATA:
27
28     def __init__(self):
29         self.host = '10.97.108.00'
30         self.client = DataFrameClient(host, '8086', 'root', 'root')
31         self.dropdb('uData')
32         self.createdb('uData')
33
34     def createdb(self, dbname):
35         print('Create database: ' + dbname)
36         self.client.create_database(dbname)
37         self.client.switch_database(dbname)
38
39     def dropdb(self, dbname):
40         print('DROP database: ' + dbname)
41         self.client.drop_database(dbname)
42
43     def dropmeas(self, measname):
44         print('DROP MEASUREMENT: ' + measname)
45         self.client.query('DROP MEASUREMENT ' + measname)
46
47     def explode(df):
48         for col in df.columns:
49             if isinstance(df.iloc[0][col], list) and col != 'neighbourCellList':
50                 df[col] = df[col].apply(lambda x: x[0])
```


Anomaly Detection Use Case (3/9)

- add RUN pip install --force-reinstall redis==3.0.1 into *ad/Dockerfile* file



```
24 ENV LD_LIBRARY_PATH /usr/local/lib:/usr/local/lib64
25 COPY local.rt /opt/route/local.rt
26 ENV RHB_SEED_RT /opt/route/local.rt
27
28 RUN apk update && apk add gcc musl-dev
29 RUN pip install ricxappframe
30 RUN pip install --force-reinstall redis==3.0.1
31
32 # install
33 COPY setup.py /tmp
34 COPY LICENSE.txt /tmp/
35 # RUN mkdir -p /tmp/ad/
36 COPY ad/ /ad
37 RUN pip install /tmp
38 ENV PYTHONUNBUFFERED 1
```

- git clone <https://github.com/ToriRobert/ad.git> -b master 後, 並從資料夾取得 *schema.json* file, 再儲存至 *ad/xapp-descriptor* 路徑
- cd ad
- docker build -t nexus3.o-ran-sc.org:10002/o-ran-sc/ric-app-ad:0.0.2 .
- cd xapp-descriptor
- export CHART_REPO_URL=http://0.0.0.0:8090
- dms_cli onboard config.json schema.json % 上架 xApp

點不能遺漏!

```
{
  "status": "Created"
}
```

- dms_cli get_charts_list % 檢查xApp是否成功上架

```
"ad": [
  {
    "apiVersion": "v1",
    "appVersion": "1.0",
    "created": "2023-08-18T11:37:20.297966861Z",
    "description": "Standard xApp Helm chart",
    "digest": "aaf0e3acfd4d3d3a8e59e8a92340a0cbb4c8499bcc2c8fe55dcb7aa57b0e2b4",
    "name": "ad",
    "urls": [
      "charts/ad-0.0.2.tgz"
    ],
    "version": "0.0.2"
  }
],
```

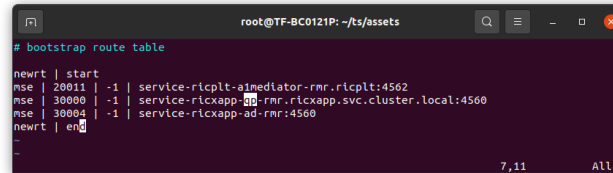
- dms_cli install ad 0.0.2 ricxapp % 安裝 xApp

```
status: OK
```

Anomaly Detection Use Case (4/9)

- Install ts xApp:

- `git clone --branch f-release https://gerrit.o-ran-sc.org/r/ric-app/ts`
- 進入 `assets/bootstrap.rt` 檔, 修改 `qpdrive` 至 `qp`



```
root@TF-BC0121P: ~/ts/assets
# bootstrap route table
newrt | start
mse | 20011 | -1 | service-ricplt-ainediator-rmr.ricplt:4562
mse | 30000 | -1 | service-ricxapp-20-rmr.ricxapp.svc.cluster.local:4560
mse | 30004 | -1 | service-ricxapp-ad-rmr:4560
newrt | end
```

- `cd ts`
- `docker build -t nexus3.o-ran-sc.org:10002/o-ran-sc/ric-app-ts:1.2.4 .`
- `cd xapp-descriptor`
- `export CHART_REPO_URL=http://0.0.0.0:8090`
- `dms_cli onboard config-file.json schema.json`

```
{
  "status": "Created"
}
```

- `dms_cli get_charts_list`

```
{
  "trafficxapp": [
    {
      "apiVersion": "v1",
      "appVersion": "1.0",
      "created": "2023-08-18T07:49:10.360429832Z",
      "description": "Standard xApp Helm Chart",
      "digest": "71caa8bc1f96f369fe9dd5dbd35b7bd47ffb9ba3e8dcdfdc91ed71c24ab2c0a4",
      "name": "trafficxapp",
      "urls": [
        "charts/trafficxapp-1.2.4.tgz"
      ],
      "version": "1.2.4"
    }
  ]
}
```

- `dms_cli install trafficxapp 1.2.4 ricxapp`

```
status: OK
```

Anomaly Detection Use Case (5/9)

- Install qp xApp:

- `git clone --branch f-release https://gerrit.o-ran-sc.org/r/ric-app/qp`
- Same as ad xApp installation, modify the host IP (influxdb service IP) in *database.py* and *insert.py* file
- Also, add RUN `pip install --force-reinstall redis==3.0.1` into *ad/Dockerfile* file
- assign 0.13.5 version for statsmodels and 1.3.5 for pandas in *qp/setup.py* file % 如右下圖
- `git clone https://github.com/ToriRobert/qp.git -b e-release` 後, 並從資料夾取得 *schema.json* file, 再儲存至 *qp/xapp-descriptor* 路徑
- `cd qp`
- `docker build -t nexus3.o-ran-sc.org:10002/o-ran-sc/ric-app-qp:0.0.4 .`
- `cd xapp-descriptor`
- `export CHART_REPO_URL=http://0.0.0.0:8090`
- `dms_cli onboard config.json schema.json`

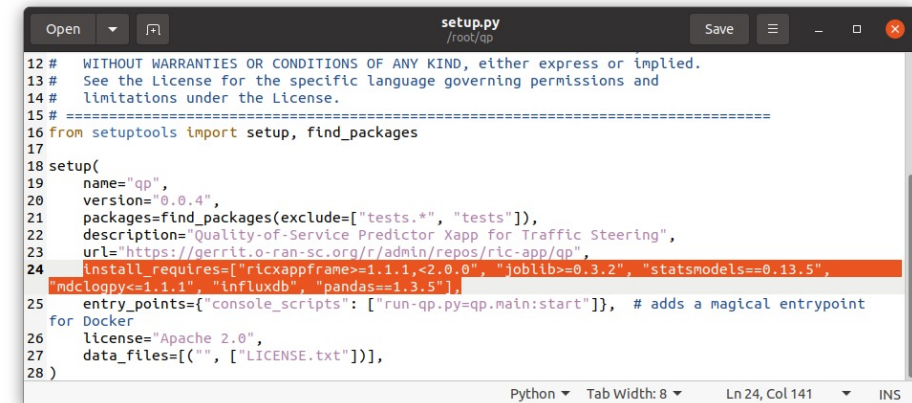
```
{
  "status": "Created"
}
```

- `dms_cli get_charts_list`

```
{
  "qp": {
    "apiVersion": "v1",
    "appVersion": "1.0",
    "created": "2023-08-18T07:50:19.041852392Z",
    "description": "Standard xApp Helix Chart",
    "digest": "d1c09bdb59ee0a5d429d4fb4d1bfba00dde452c7f204ee743f3ba855994d97e",
    "name": "qp",
    "urls": [
      "charts/qp-0.0.4.tgz"
    ],
    "version": "0.0.4"
  }
}
```

- `dms_cli install qp 0.0.4 ricxapp`

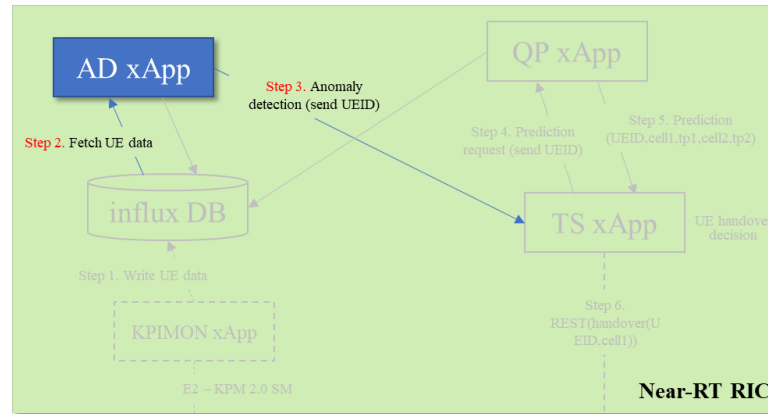
```
status: OK
```



```
12 # WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
13 # See the License for the specific language governing permissions and
14 # limitations under the License.
15 # =====
16 from setuptools import setup, find_packages
17
18 setup(
19     name="qp",
20     version="0.0.4",
21     packages=find_packages(exclude=["tests.*", "tests"]),
22     description="Quality-of-Service Predictor Xapp for Traffic Steering",
23     url="https://gerrit.o-ran-sc.org/r/admin/repos/ric-app/qp",
24     install_requires=["ricxappframe==1.1.1", "influxdb", "pandas==1.3.5", "statsmodels==0.13.5", "joblib==0.3.2"],
25     entry_points={ "console_scripts": ["run-qp.py=qp.main:start"] }, # adds a magical entrypoint
26     for Docker
27     license="Apache 2.0",
28     data_files=[("", ["LICENSE.txt"])],
29 )
```

Anomaly Detection Use Case (6/9)

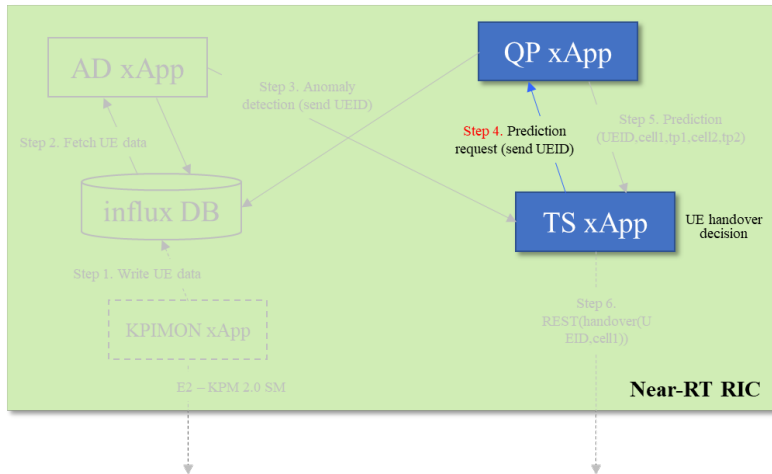
- Check for logs in xApp:
 - `kubectl logs -f <xApp pod name> -n ricxapp`
 - `kubectl get pods -n ricxapp` % check xApp status and pod name
- AD xApp logs:
 - AD xApp 成功將異常 UE 的資訊傳送給 TS xApp。



```
root@TF-BC0121P: ~/qp/xapp-descriptor
/opt/conda/lib/python3.9/site-packages/sklearn/base.py:464: UserWarning: X does not have valid feature names, but IsolationForest was fitted with feature names
warnings.warn(
[INFO] Sending Anomalous UE to TS
[INFO] Message to TS: message sent Successfully
[INFO] Received acknowledgement from TS (TS_ANOMALY_ACK): {'payload': b'[{"du-id": 1001, "ue-id": "Car-4", "measTimeStampRf": 1692365956956, "Degradation": "RSRP RSSINR"}]', 'payload length': 99, 'message type': 30004, 'subscription id': -1, 'transaction id': b'26d711d83dcd11eeb6a6225f86831d4', 'message state': 0, 'message status': 'RMR_OK', 'payload max size': 3136, 'meid': b'', 'message source': 'service-ricxapp-trafficxapp-rmr.ricxapp:4560', 'errno': 0}
/opt/conda/lib/python3.9/site-packages/sklearn/base.py:464: UserWarning: X does not have valid feature names, but IsolationForest was fitted with feature names
warnings.warn(
/opt/conda/lib/python3.9/site-packages/sklearn/base.py:464: UserWarning: X does not have valid feature names, but IsolationForest was fitted with feature names
```

Anomaly Detection Use Case (7/9)

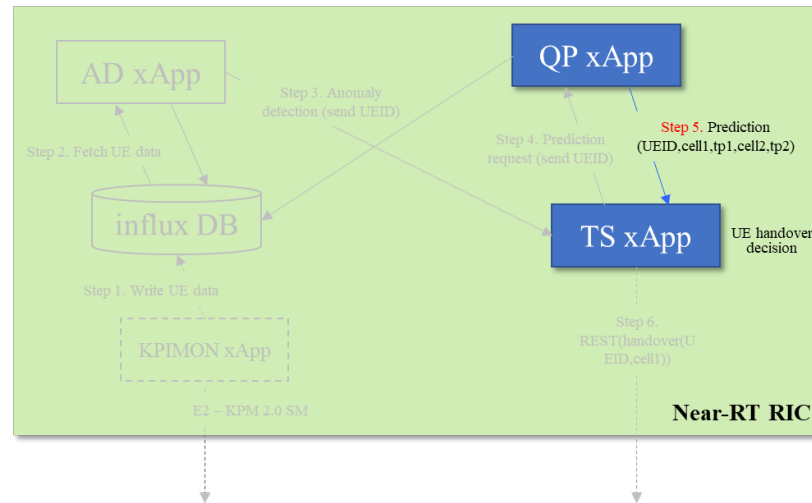
- TS xApp logs:
 - TS xApp 收到 AD xApp 傳回的異常 UE 資訊。
 - TS xApp 向 QP xApp 要求對 UE 進行 QoE 的預測。



```
root@TF-BC0121P: ~/qp/xapp-descriptor
unable to complete the request url=http://127.0.0.1:5000/api/echo error=Failed to connect to 127.0.0.1 port 5000: Connection refused
[ERROR] Unexpected HTTP code 0 from http://127.0.0.1:5000/api/echo
[ERROR] HTTP payload is
[INFO] AD Callback got a message, type=30003, length=99
[INFO] Payload is [{"du-id": 1001, "ue-id": "Car-2", "measTimeStampRf": 1692365982596, "Degradation": "RSRP RSSINR"}]
[INFO] Prediction Request length=30, payload={"UEPredictionSet": ["Car-2"]}
[INFO] Prediction Callback got a message, type=30002, length=142
[INFO] Payload is {"Car-4": {"c1/B13": [2656, 2656], "c1/B2": [2, 2], "c1/N77": [0, 0], "c4/B13": [3421, 3421], "c2/B13": [4118, 4118], "c3/B13": [1063, 1063]}}
[INFO] Sending a HandOff CONTROL message to "http://127.0.0.1:5000/api/echo"
[INFO] HandOff request is {
  "command": "HandOff",
  "seqNo": 415,
  "ue": "Car-4",
  "fromCell": "c1/B13",
  "toCell": "c2/B13",
  "timestamp": "Fri Aug 18 13:45:52 2023",
  "reason": "HandOff Control Request from TS xApp",
  "ttl": 10
}
[ERROR] Unexpected HTTP code 0 from http://127.0.0.1:5000/api/echo
[ERROR] HTTP payload is
unable to complete the request url=http://127.0.0.1:5000/api/echo error=Failed to connect to 127.0.0.1 port 5000: Connection refused
[INFO] AD Callback got a message, type=30003, length=99
```

Anomaly Detection Use Case (8/9)

- QP xApp logs:
 - QP xApp 接收到來自 TS xApp 的異常 UE 資訊。
 - QP xApp 成功將異常 UE 的 QoE 預測資訊回傳給 TS xApp。

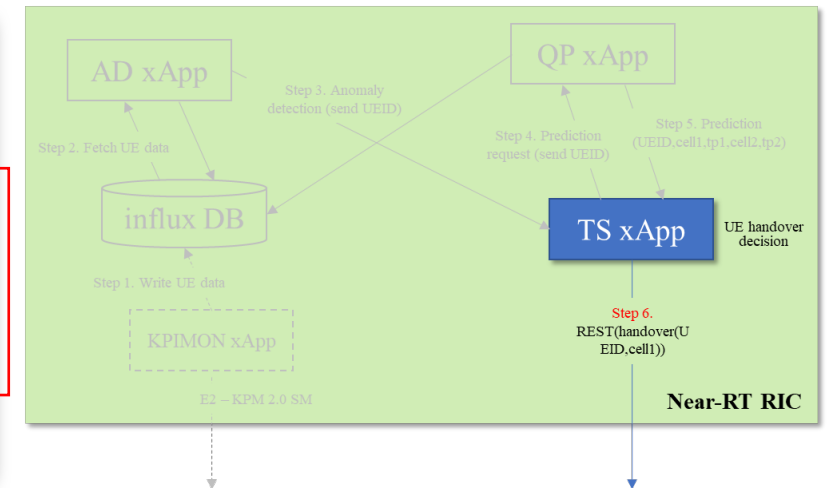


```
root@TF-BC0121P: ~/qp/xapp-descriptor
{"ts": 1692421781780, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "predict handler received payload b'{'UEPredictionSet\": [\"Car-2\"]}'"}
{"ts": 1692421782246, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "Sending message to ts : {'Car-2\": {'c1/B13\": [2656, 2656], \"c1/B2\": [2, 2], \"c1/N77\": [0, 0], \"c4/B13\": [3421, 3421], \"c5/B13\": [2659, 2659], \"c6/B13\": [0, 0]}}"}
{"ts": 1692421782246, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "predict handler: sent message successfully"}
{"ts": 1692421782246, "crit": "DEBUG", "id": "ricxappframe.xapp_frame", "mdc": {}, "msg": "run: invoking msg handler on type 30000"}
{"ts": 1692421782246, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "predict handler received payload b'{'UEPredictionSet\": [\"Car-4\"]}'"}
{"ts": 1692421782668, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "Sending message to ts : {'Car-4\": {'c1/B13\": [2656, 2656], \"c1/B2\": [2, 2], \"c1/N77\": [0, 0], \"c4/B13\": [3421, 3421], \"c2/B13\": [4118, 4118], \"c3/B13\": [1063, 1063]}}"}
{"ts": 1692421782668, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "predict handler: sent message successfully"}
{"ts": 1692421782668, "crit": "DEBUG", "id": "ricxappframe.xapp_frame", "mdc": {}, "msg": "run: invoking msg handler on type 30000"}
{"ts": 1692421782668, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "predict handler received payload b'{'UEPredictionSet\": [\"Pedestrian-1\"]}'"}
{"ts": 1692421783051, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "Sending message to ts : {'Pedestrian-1\": {'c1/B13\": [2656, 2656], \"c1/B2\": [2, 2], \"c1/N77\": [0, 0], \"c4/B13\": [3421, 3421], \"c2/B13\": [4118, 4118], \"c3/B13\": [1063, 1063]}}"}
{"ts": 1692421783051, "crit": "DEBUG", "id": "qp.main", "mdc": {}, "msg": "predict handler: sent message successfully"}
```

Anomaly Detection Use Case (9/9)

- TS xApp logs:
 - TS xApp 收到 QP xApp 傳回的 UE QoE 預測資訊。
 - TS xApp 進行 UE handoff 決策，並將 handoff 要求訊息傳送至指定的位置。

```
root@TF-BC0121P: ~/qp/xapp-descriptor
unable to complete the request url=http://127.0.0.1:5000/api/echo error=Failed to connect to 127.0.0.1 port 5000: Connection refused
[ERROR] Unexpected HTTP code 0 from http://127.0.0.1:5000/api/echo
[ERROR] HTTP payload is
[INFO] AD Callback got a message, type=30003, length=99
[INFO] Payload is [{"du-id": 1001, "ue-id": "Car-2", "measTimeStampRf": 1692365982596, "Degradation": "RSRP RSSINR"}]
[INFO] Prediction Request length=30, payload={"UEPredictionSet": [{"Car-2"}]}
[INFO] Prediction Callback got a message, type=30002, length=142
[INFO] Payload is [{"Car-4": {"c1/B13": [2656, 2656], "c1/B2": [2, 2], "c1/N77": [0, 0], "c4/B13": [3421, 3421], "c2/B13": [4118, 4118], "c3/B13": [1063, 1063]}}]
[INFO] Sending a HandOff CONTROL message to "http://127.0.0.1:5000/api/echo"
[INFO] Handoff request is {
  "command": "HandOff",
  "seqNo": 415,
  "ue": "Car-4",
  "fromCell": "c1/B13",
  "toCell": "c2/B13",
  "timestamp": "Fri Aug 18 13:45:52 2023",
  "reason": "HandOff Control Request from TS xApp",
  "ttl": 10
}
[ERROR] Unexpected HTTP code 0 from http://127.0.0.1:5000/api/echo
[ERROR] HTTP payload is
unable to complete the request url=http://127.0.0.1:5000/api/echo error=Failed to connect to 127.0.0.1 port 5000: Connection refused
[INFO] AD Callback got a message, type=30003, length=99
```



Outline

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- O-RAN implementation guide on ubuntu 20.04
 - Install k8s, helm, and docker
 - Install Near-RT RIC
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 - Anomaly Detection Use Case
- **Reference**

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